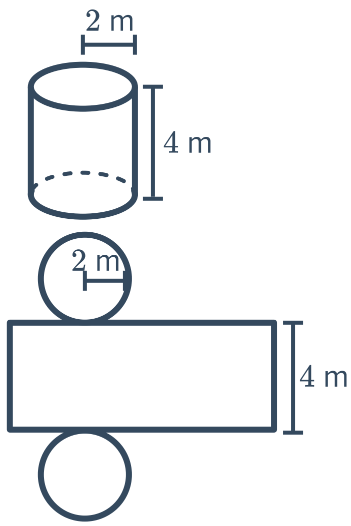
Surface area of cylinders

1 Consider the following cylinder together with its net:

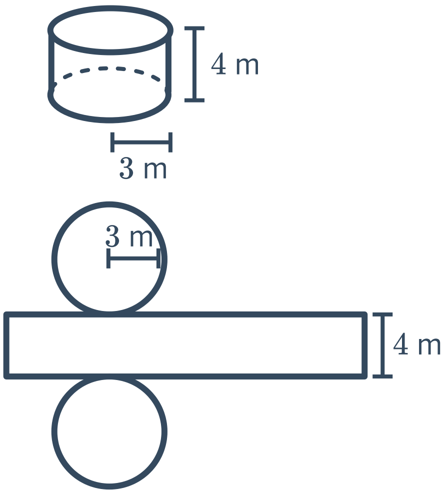
a Find the circumference of the circular base. Round your answers to two decimal places.

b Hence, find the area of the curved face of the cylinder. Round your answers to two decimal places.

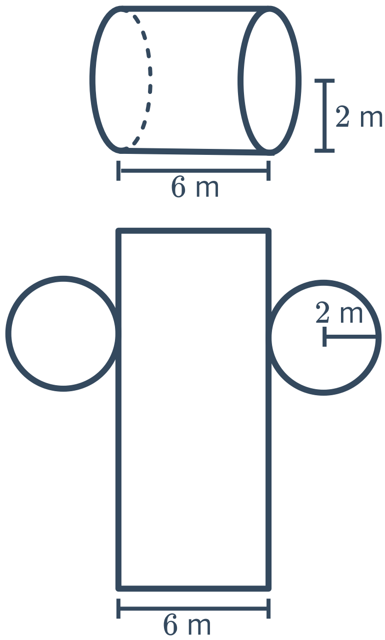


2 For each of the following pair of cylinders and their nets, find the curved surface area of the cylinder. Round your answers to two decimal places.

a



b

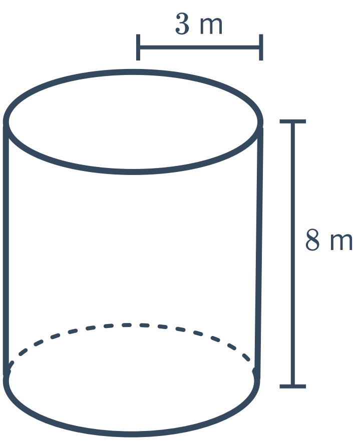


3 For each of the following cylinders:

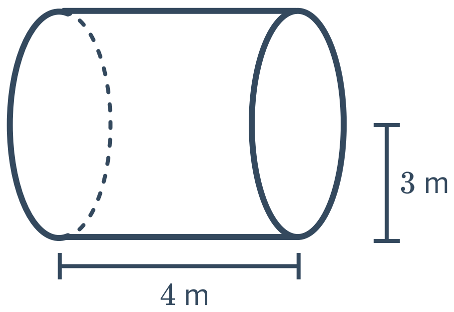
i Find the curved surface area of the cylinder. Round your answers to two decimal places.

ii Find the total surface area of the cylinder. Round your answers to two decimal places.

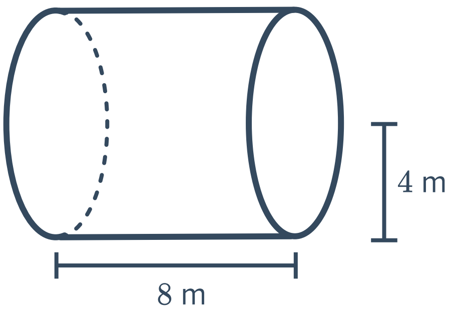
a



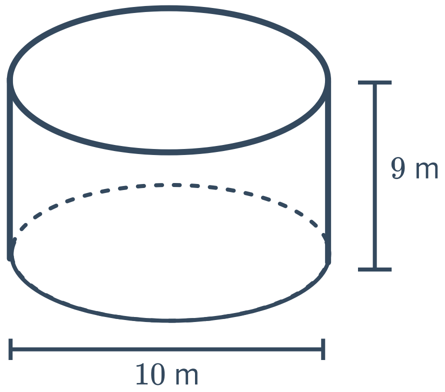
b



c



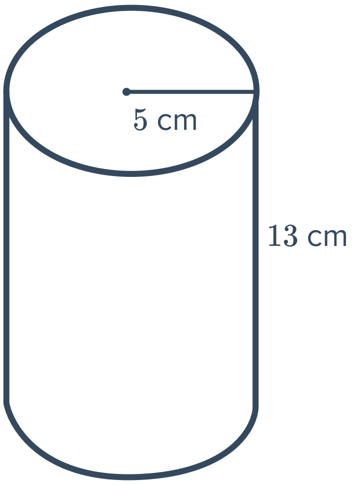
d



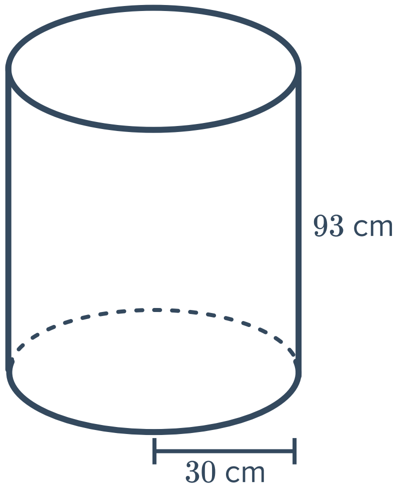
4 Find the surface area of the following cylinders.

Round your answers to two decimal places.

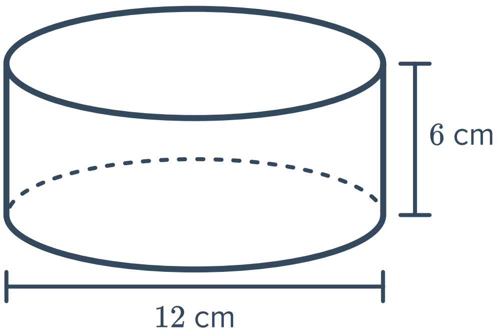
a



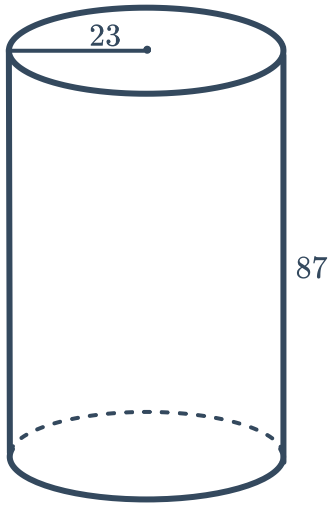
b



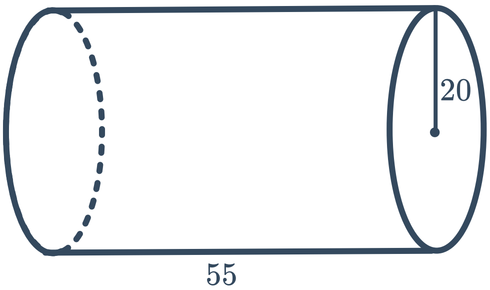
c



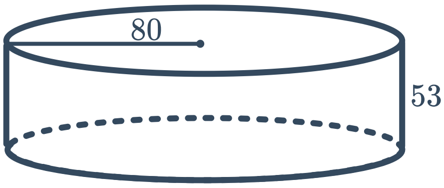
d



e



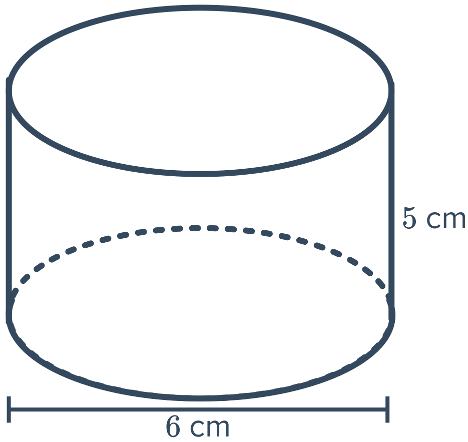
f



5 Consider the following cylinder:

a Find the surface area of the cylinder to one decimal place.

b Find the surface area of the cylinder in square millimetres to the nearest whole number.



6 The area of the circular face on a cylinder is $8281\pi\ \mathrm{m}^2$. The total surface area of the cylinder is $25\,662\pi\ \mathrm{m}^2$.

a Find the radius of the cylinder.

b Find the height of the cylinder.

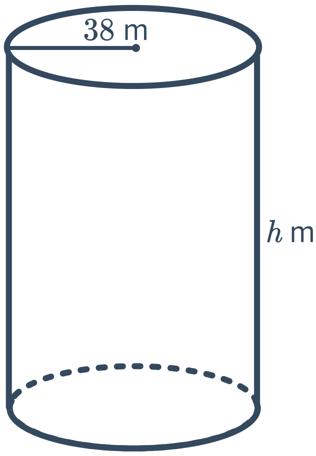
7 The area of the circular face on a cylinder is $225\pi\ \mathrm{m}^2$. The total surface area of the cylinder is $3210\pi\ \mathrm{m}^2$.

a Find the value of the radius of the cylinder.

b Find the height of the cylinder.

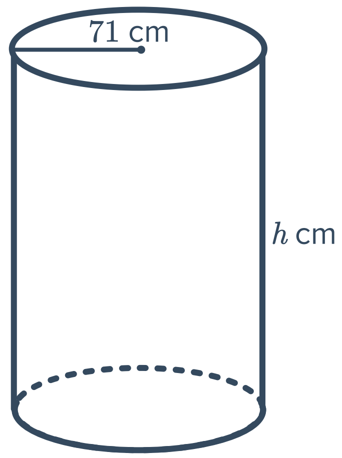
8 The following cylinder has a surface area of $25\,547\ \mathrm{m}^2$:

Find the height the cylinder. Round your answer to the nearest whole number.

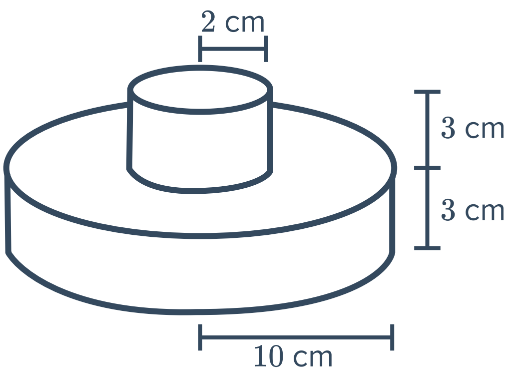


9 The following cylinder has a surface area [redacted] $54\,425\ \text{cm}^2$:

Find the height the cylinder. Round your answer to the nearest whole number.



10 Find the surface area of the following figure.



11 A cylindrical can of radius $7\ \text{cm}$ and height $10\ \text{cm}$ is open at one end. What is the external surface area of the can? Round your answer to two decimal places.

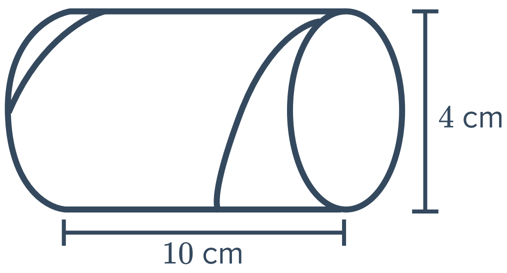
12 Find the exact surface area of a cylinder with diameter $6\ \text{cm}$ and height $21\ \text{cm}$ by leaving your answer in terms of π.

13 Find the lateral surface area of a cylinder with a radius of $9.2\ \text{m}$ and a height of $15.1\ \text{m}$. Round your answer to one decimal places.

Applications

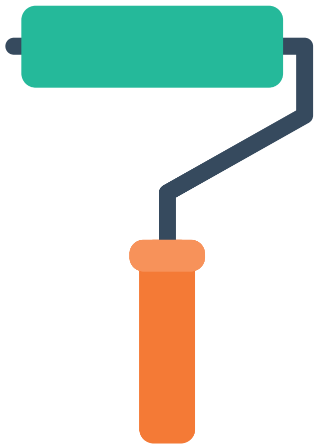
14 Paul is using a toilet paper roll for crafts. He has measured the toilet paper roll to have a diameter 4 cm and a length 10 cm.

Find the surface area of the toilet paper roll. Round your answer to two decimal places.



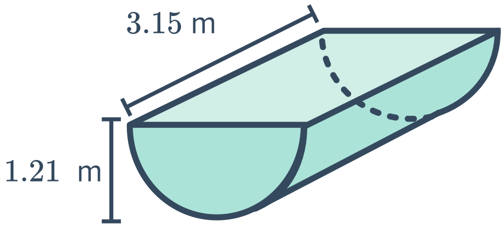
15 A paint roller is cylindrical in shape. It has a diameter of 6.2 cm and a width of 38.1 cm.

Find the area painted by the roller when it makes one revolution. Round your answer to two decimal places.



16 The diagram shows a water trough in the shape of a half cylinder:

Find the exact surface area of the outside of this water trough, leaving your answer in terms of π.



17 Amy and Vincent each have a cylinder. Amy's cylinder has a diameter of 8 cm and a height of 9 cm. Vincent's cylinder has a diameter of 9 cm and a height of 8 cm.

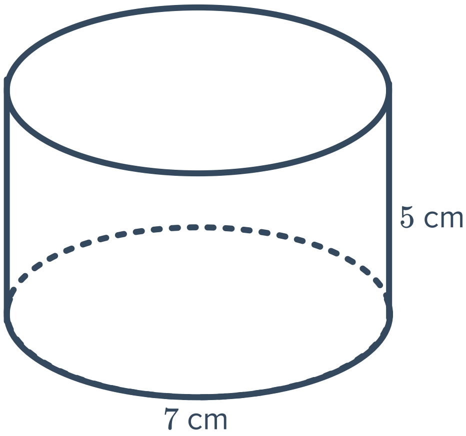
a Find the surface area of Amy's cylinder. Round your answer to two decimal places.

b Find the surface area of Vincent's cylinder.

c Which cylinder has a larger surface area? Round your answer to two decimal places.

18 Jenny wants to make several cans like the one shown. She plans to cut them out of a sheet of material that has an area of 1683 cm^2.

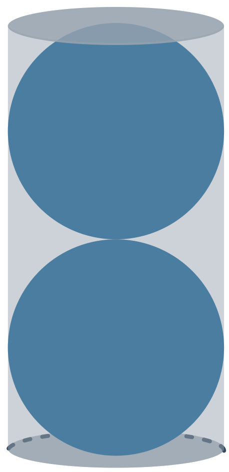
How many complete cans can she make?



19 If a spherical ball with a radius of 4.9 m fits exactly inside a cylinder, find the surface area of the cylinder.

20 The two identical spherical balls with radii of 2.6 cm fit exactly inside a cylinder.

Find the surface area of the closed cylinder.



21 Find the surface area of the brickwork for this silo. Assume that there is a brick roof and no floor.